

Wharton competition preparation package

Working copy - blocked for submission. AI-assisted material, 2026-09-27.

This package applies the Wharton megaprompt to the actual Global High School Investment Competition. The product is a traceable investment decision process and supporting research, not a venture with invented revenue or customers. It supersedes older competition drafts for current preparation. Historical PDFs remain preserved and must not be submitted.

Start with EXECUTIVE_BRIEF.md, then WRITTEN_RESPONSES.md, PITCH_CONTENT.md, TIMED_PITCH.md, DEMO_SCRIPT.md, JUDGE_QA.md, and EVIDENCE_APPENDIX.md. FINANCIAL_STRESS.md is generated from retained calculator outputs. DECISION_ROOM.html is an offline interactive view of those same outputs; it places no trades. IPS_DRAFT.md and output/pdf/WHARTON_IPS_DRAFT_BLOCKED.pdf provide a source-formatted, blocked IPS draft. output/pdf/WHARTON_EXECUTIVE_BRIEF.pdf is the one-page brief; output/pdf/WHARTON_PREPARATION_PACK.pdf is the complete internal reading copy (paths relative to repository root).

No client identity, wealth, objective, risk tolerance, eligibility, team member, execution, or claimed student experience is filled in from inference. The current public calendar is in protocol/OFFICIAL_RULES_NOTE.md. IPS and Trading Notes instructions were recovered from Downloads and are mapped in protocol/VERIFIED_DELIVERABLE_REQUIREMENTS.md. Other Apply materials remain missing.

Release gates

1. Obtain the current case study, trading rules, final-report instructions and any updates from the registered team's Apply account; preserve each file and hash.
2. Extract client and eligibility facts with source file/page references. The source data should distinguish student roster from client beneficiaries; the old CLIENT_CONSTRAINTS.yaml roster section describes beneficiaries, not team eligibility.
3. Verify eligible instruments and real WInS records. The recovered Trading Notes PDF confirms three executed notes, exact original wording and reflections of at most 100 words each.
4. Students review, make and defend their own decisions, replace missing facts, and attribute any AI material used according to official policy. They must not describe AI-generated analysis as their own experience.
5. Check final documents against the actual rubric and rendered limits; obtain explicit submission authority.

These gates do not prevent using the local research and calculator now. They prevent calling a draft final or placing trades without authority.

Written working responses

Not a submitted application or final IPS. IPS and Trading Notes instructions were recovered; the complete client case, final-report instructions and numerical rubric remain missing. The following complete narrative blocks answer the known preparation topics, not an invented official form. AI-assisted material requires student verification and attribution. No fabricated personal or teamwork narrative is included.

Strategy summary

Our proposed decision process begins with the client's documented objectives and obligations. It separates cash needed for payments from capital that can bear investment risk. We compare transparent portfolio rules against a fully invested reference, retain downside failures, and record the purpose and evidence behind each security decision. We will not rely on the laboratory's earlier forecasting claim: it was withdrawn after leakage was found. A portfolio recommendation remains conditional on the actual client case, eligible instruments and risk constraints.

Problem definition and target user

The immediate user is the student investment team preparing a client-aligned recommendation. Its problem is deciding what can be supported by evidence, not maximizing a research scoreboard. The beneficiary is the competition client once the official case is available. The workspace establishes only the first name Laura and the broad operating/facility objectives from the recovered instructions; it cannot establish complete identity, horizon, withdrawals or risk tolerance. The recovered IPS instructions confirm the operating-commitment and facility-contribution objectives. The complete client case is still needed for their amounts, timing and constraints.

Research and differentiated value

The laboratory contains twelve implemented models, but implementation is distinct from investment usefulness. An integrity audit invalidated the synthetic forecasting result, identified overlapping experimental folds, and corrected the interpretation of a cached ETF target. The repaired study compares causal features with simple baselines and retains null results. The proposed value is a decision record that can be checked: each client constraint, input assumption and quantitative statement is linked to a source. We make no claim that this process is unique or commercially validated.

Alternatives and competitive landscape

Approach	Potential use	Main weakness	Evidence needed before adoption
Static diversified allocation	Transparent reference	May not match obligations or risk capacity	Client case, eligibility and downside analysis
Fixed cash reserve	Simple liquidity buffer	Idle cash has opportunity cost; buffer may be mis-sized	Payment schedule and replenishment rule
Liability-sized reserve	Links liquidity to near-term payments	Cannot solve insolvency; depends on correct obligations	Dated obligations and inflation assumptions
EWMA risk monitoring	Tracks recent covariance mechanically	Not a validated allocation improvement here	Causal out-of-sample covariance and decision tests
Return-forecast strategy	Could adapt to genuine predictability	No supported edge in tested baseline study	Fresh data, strong comparators, costs and multiplicity control
Complex MPC / M12 controller	Intended multi-period planning	Known accounting / evidence defects	Separate repaired protocol and validated accounting

These are decision alternatives, not a priced competitor or market-share survey. No competitor revenue, total addressable market, acquisition cost or commercial traction has been measured.

Economics and financial assumptions

Venture revenue and customer unit economics are not applicable to this competition. The relevant economics are asset returns, liquidity, trading costs, inflation and obligations. The calculator normalizes starting wealth to one, models hypothetical sleeves and tests explicitly assumed return paths. It records transaction costs, payments, unpaid obligations and remaining purchasing power. It does not estimate client wealth, expected market returns, security fees, taxes or real execution costs. Costs and reserve policy must be recalibrated from the actual case and applicable trading rules before making a recommendation.

Implementation and adoption

First, students ingest the Apply materials and annotate source facts. Next, they set an eligible universe and justify a strategic allocation with an explicit benchmark. They test obligations and downside risk, then decide whether any additional model improves the decision enough to justify its complexity. Once authorized within the competition, they record WInS decisions and exact notes as executed, review deviations, and use the final report to explain what changed and why. No outreach to the competition client is proposed.

Impact and success measures

Proposed measures are source completeness, percentage of holdings with a documented client rationale, funded obligations under stated assumptions, risk and cost exposure, and traceable changes in reasoning. These are proposed tracking measures, not achieved outcomes. Simulator returns, client outcomes and competition placement are unobserved. A funding stress failure is a reason to reconsider spending, contributions, horizon or risk, not to choose a more flattering scenario.

Risk and governance

Principal risks are missing or misread client facts, unsupported return assumptions, illiquid or ineligible instruments, inflation, losses near withdrawals, model leakage, transaction costs and overstatement of student authorship. Required controls are source references, an explicit information clock, comparisons with simple baselines, preserved negative results, a decision log and student review. No client funding guarantee is made.

Client-specific sections that cannot honestly be finalized

The final IPS must state sourced goals, time horizon, wealth, payment schedule, loss capacity, liquidity need, constraints, benchmark, eligible allocation and review triggers. Trading analysis requires genuine WInS executions and the original notes. Team and school responses require real registration, roster and school documentation. None is available here; these sections are blocked rather than filled with invented facts. The recovered PDFs establish IPS and Trading Notes requirements; see protocol/VERIFIED_DELIVERABLE_REQUIREMENTS.md. Final-report requirements remain unverified.

Pitch deck content

Internal rehearsal content, not a verified competition slide requirement. Eight slides are an editorial choice. AI-assisted draft; students must validate and adapt it to their actual analysis. Keep the blocked status visible until source facts and decisions exist.

Slide 1 - Evidence before allocation

Headline: A client-aligned strategy starts with what we can defend.

On slide: Source the obligations. Compare transparent choices. Keep failed tests visible.

Visual: Three-step process, with client facts shown as a pending input. Do not use a fabricated client photo or logo endorsement.

Speaker point: This is a proposed process and research package, not a funded client portfolio.

Slide 2 - The decision to make

Headline: Liquidity needs determine how much risk is affordable.

On slide: What must be paid? When? What capital can remain invested? What losses are tolerable?

Visual: Empty source checklist labelled 'Awaiting official case'; no assumed client balances.

Speaker point: Do not turn a hypothetical calculator input into a client fact.

Slide 3 - What the research changed

Headline: An apparent forecasting win did not survive an integrity audit.

On slide: Legacy leakage discovered; claim withdrawn; repaired baseline results do not establish an edge over the historical mean.

Visual: Existing reports/figures/repairs_oos_r2.png with both panels and labels intact. Explain that the high leaky-feature result is invalid evidence.

Speaker point: This result limits use of the tested forecasts. It does not validate every alternative.

Slide 4 - Transparent alternatives

Headline: Complexity must earn its place.

On slide: Fully invested reference; fixed cash reserve; liability-sized reserve. EWMA is a monitoring candidate, not a proven improvement.

Visual: Three-column policy comparison using WRITTEN_RESPONSES.md.

Speaker point: No exact portfolio weights are client recommendations yet.

Slide 5 - Downside before upside

Headline: Test whether payments remain affordable.

On slide: Joint asset loss; adverse inflation; heavier payments; early versus late loss; costs.

Visual: Calculator output at 10 bps, with the assumption-only label and insolvency path retained.

Speaker point: Hand-designed paths answer 'what if', not 'how likely'. Do not describe 75 combinations as 75 independent market observations.

Slide 6 - Decision and execution record

Headline: Each decision needs a reason and a source.

On slide: Client constraint -> eligible holding -> risk and liquidity check -> student decision -> WInS evidence -> reflection.

Visual: Process sequence; execution stage visibly pending.

Speaker point: There are no verified WInS trades in this package.

Slide 7 - What can fail

Headline: Reserves do not make an unaffordable plan solvent.

On slide: Spending shocks; persistent inflation; correlated losses; opportunity cost; model errors.

Visual: Unpaid obligations and real remaining wealth side by side from FINANCIAL_STRESS.md.

Speaker point: A more complex controller is not a substitute for sound accounting or realistic liabilities.

Slide 8 - Conditions for a recommendation

Headline: Finish the source-to-decision chain.

On slide: Verify case and rules; set constraints; justify allocation; record real decisions; check final materials.

Speaker point: The current claim is that the preparation tools and evidence are reviewable. The client recommendation remains blocked.

Footer for all slides: AI-assisted working copy; source register in EVIDENCE_APPENDIX.md. Official presentation duration and format unverified.

Timed pitch script

Five-minute rehearsal plan, not an official time limit. Approximately three minutes of spoken material plus two minutes to show evidence and pause. Actual duration must be measured by the student presenters; no rehearsal has been claimed. AI-assisted draft, not a student's personal account.

0:00-0:40 - Purpose

The question is not which model produces the most impressive chart. It is which investment decisions can be defended for the client's actual goals. Our working approach separates payment needs from risk-bearing capital, compares transparent choices, and makes uncertainty visible. The client case is not yet in this research workspace, so no final allocation or client outcome is claimed.

0:40-1:20 - Evidence

The laboratory's earlier synthetic forecasting win was withdrawn after leakage was discovered. Its features contained information about the return being predicted. The repaired baseline study did not establish a statistically supported advantage over the historical mean. That is a reason to avoid relying on the tested forecasts. It is not proof that a different allocation method is optimal. Complex models and a frozen controller also have unresolved evidence problems.

1:20-2:10 - Proposed process

The first step is to extract the client's goals, payment dates, constraints and loss capacity from the official case. The next is to compare a simple invested reference, a fixed cash buffer and a reserve linked to obligations. Cash can reduce exposure near payments, but it also has an opportunity cost. Eligible instruments, portfolio weights and review triggers will need a client-specific rationale before any recommendation.

2:10-3:20 - Show the stress tool

This calculator starts with normalized wealth and clearly labelled assumptions. It applies costs to trades, earns the specified returns, and pays each obligation once. It keeps unpaid obligations visible and never supplies hidden rescue capital. Here are an early joint loss, high inflation and an unaffordable spending case. These are deliberately chosen sensitivities, not historical performance or estimated probabilities. Compare payments and remaining purchasing power together.

Presenter action: spend the remaining time in this interval switching between the steady and funding-failure scenarios at 10 bps. Point to the assumption label and first missed payment. Do not hide the failure.

3:20-4:20 - Governance

Before implementation, the team must verify the current case, eligible universe and trading rules. Students must make and understand their own decisions. Every executed note needs its original WInS record. The final explanation should connect each holding to a sourced objective and distinguish what the team decided from AI-assisted research. There are no real executions or fabricated teamwork stories in this package.

Presenter action: show the evidence appendix and source checklist during the remaining time.

4:20-5:00 - Close

The preparation work gives us a traceable starting point and a way to challenge assumptions. It does not supply a client funding guarantee or an investment edge. The remaining task is to turn verified client facts into a justified, eligible portfolio, preserve the actual decision history, and submit only materials the students can explain and defend.

Local decision-tool demonstration

A demonstration of real local calculation and saved evidence, not simulated customer traction or executed WInS trades. No login, network service or trading permission is needed.

1. Run `PYTHONPATH=src .venv/bin/python scripts/run_competition_stress.py`, then `.venv/bin/python scripts/build_competition_artifacts.py`. Open `reports/competition_package/DECISION_ROOM.html` locally.
2. Read the banner: all inputs are assumptions, initial wealth is normalized, and no portfolio is approved for the client. Show the visible source hashes and config path.
3. Select `steady_assumption`, `liability_reserve`, 10 bps. Read end wealth, funded share, unpaid balance and transaction costs. These values come from the retained JSON trajectory.
4. Switch to `early_joint_crash`, then `late_joint_crash` without changing policy or cost. Explain the payment timing effect; do not call either path a forecast.
5. Select `funding_failure`. Show the first missed payment and unpaid balance. Switch policies and retain the adverse finding. Cash changes exposure; it does not eliminate an infeasible obligation schedule.
6. Change costs from 0 to 50 bps. Explain that the values are cost sensitivities, not verified execution costs. Cash interest and taxes are omitted assumptions.
7. Open `runs/competition_stress/SUMMARY.csv` or the JSON and cross-check the displayed combination. The complete monthly ledger records starting wealth, trades, fees, payment and ending wealth.
8. Open `EVIDENCE_APPENDIX.md`. Explain why legacy model results and the old synthetic win cannot be used as support. Show the client-source and trading-record blockers.

Expected outcome: the user can trace a displayed number to a deterministic calculation and identify its limits. No result establishes client suitability, empirical superiority, or submission readiness.

Fallback: If interactive scripting is unavailable, use `FINANCIAL_STRESS.md` and the JSON trajectories. Do not replace missing outputs with static invented numbers.

Financial sensitivity results

ASSUMPTION ONLY. Not market history, client wealth, a forecast, a probability estimate, or WInS performance.

Generated from configs/competition_stress.json by scripts/run_competition_stress.py. Units: initial wealth = 1. No deposits, taxes, income, financing or cash yield. Two hypothetical sleeves; monthly rebalancing and pro-rata liquidation. Asset returns and inflation are explicit scenario assumptions. Costs apply to every risky-asset purchase and sale, including forced sales for payments.

Primary display: 10 bps. Full sensitivity: 0/5/10/25/50 bps in SUMMARY.csv. These costs are research assumptions, not verified WInS rules. Reserve policy uses a fixed 2.5% planning inflation rate even when realized scenario inflation is higher. Future asset returns never enter policy decisions.

Scenario	Policy	End wealth	Paid / due	Unpaid	First miss
steady_assumption	fully_invested	1.0275	100.00%	0.0000	none
steady_assumption	fixed_cash_20pct	0.9601	100.00%	0.0000	none
steady_assumption	liability_reserve	1.0187	100.00%	0.0000	none
early_joint_crash	fully_invested	0.6560	100.00%	0.0000	none
early_joint_crash	fixed_cash_20pct	0.6797	100.00%	0.0000	none
early_joint_crash	liability_reserve	0.6561	100.00%	0.0000	none
late_joint_crash	fully_invested	0.7424	100.00%	0.0000	none
late_joint_crash	fixed_cash_20pct	0.7469	100.00%	0.0000	none
late_joint_crash	liability_reserve	0.7368	100.00%	0.0000	none
stagnation_high_inflation	fully_invested	0.2778	100.00%	0.0000	none
stagnation_high_inflation	fixed_cash_20pct	0.2781	100.00%	0.0000	none
stagnation_high_inflation	liability_reserve	0.2779	100.00%	0.0000	none
funding_failure	fully_invested	0.0000	61.47%	0.5552	84
funding_failure	fixed_cash_20pct	0.0000	62.89%	0.5348	86
funding_failure	liability_reserve	0.0000	63.10%	0.5318	86

Interpretation limits

A reserve can soften a near-term market shock but cannot make an infeasible spending plan solvent. In high-inflation or large-payment scenarios, compare unpaid obligations before terminal wealth: a portfolio ending with more wealth may simply have paid less. Net drawdown excludes withdrawals; end wealth includes them. The reserve is held as zero-yield cash, so its opportunity cost is explicit. The fixed-cash and liability policies do not isolate cash level from reserve timing; no causal superiority claim follows.

Early and late shocks use the same shock vector and otherwise identical return assumptions. They illustrate sequence-of-returns sensitivity. Five hand-designed paths are not a calibrated return distribution: do not attach confidence intervals or failure probabilities to them. EWMA sizing, M11 MPC, and M12 are not implemented in this calculator. The calculator tests accounting and transparent policy assumptions only.

Sixty hard judge questions and defensible answers

Internal practice, not official questions or a claim of student experience. Answers reflect the actual workspace on 2026-09-27. Students should replace conditional answers only with verified facts. Evidence IDs refer to EVIDENCE_APPENDIX.md.

Client and suitability

01. Who exactly is the client?

The recovered Trading Notes instructions name Laura, but the full current case and identity are missing. We cannot substitute a prior-season client. This blocks a final recommendation. [S04, S06]

02. What is the client's investment objective?

The recovered instructions establish an operating commitment, a possible facility contribution and financial flexibility. Their amounts, dates and constraints are missing. The normalized stress inputs remain assumptions. [S05-S06]

03. How much wealth is available?

No sourced client balance is available. A starting value of one in the calculator is a unit convention. It must not be quoted as actual money or converted into an invented account balance.

04. How did you choose the horizon?

The calculator uses 120 months to illustrate cash-flow effects. It is a scenario assumption, not a sourced client horizon. The final IPS must use the actual horizon and payment dates.

05. How do you distinguish risk tolerance from risk capacity?

Tolerance concerns willingness to endure losses; capacity concerns whether losses jeopardize obligations. Neither can be filled from model output. Both require the case, and conflicts must be explained before allocating risk.

06. Why not put everything in cash?

Cash reduces exposure to asset losses but can lose purchasing power and forgo growth. Our zero-yield cash assumption makes that tradeoff explicit. Whether it is suitable depends on the client's constraints and available instruments.

07. What is the minimum reserve?

No client reserve has been approved. The calculator's candidate rule covers up to twelve modelled payments using a fixed planning inflation assumption. A real rule needs the actual schedule and liquidity constraints.

08. What happens if obligations exceed assets?

The calculator records unpaid obligations and the first shortfall instead of borrowing or adding hidden capital. A real plan would need an explicit reconsideration of spending, contributions, horizon or risk; returns cannot be guaranteed. [C06]

09. Which securities are eligible?

The current team-gated trading rules are missing. Hypothetical asset sleeves in the calculator do not establish any ticker's eligibility. No security has been recommended on that basis.

10. Did you interview the client?

No. No interview is claimed, and the public rules prohibit contacting the competition client. The official case is the proper source. [S02]

Strategy and alternatives

11. What is actually distinctive about this approach?

Its intended strength is traceability from client fact to assumption to decision. We have not established commercial novelty or superiority over other teams. The preserved correction of an invalid result is evidence of transparency, not a competitive ranking.

12. Why use machine learning at all?

It is a research candidate, not a requirement. Tested baseline forecasts do not establish a reliable edge, so the current decision process does not depend on them. Any added model must justify its complexity with valid evidence. [C03]

13. Why not use a static diversified reference?

That is an important comparator. It may be adequate, but must still match the client's obligations, loss capacity and eligible universe. The calculator includes a simple fully invested reference without declaring it suitable.

14. Is a fixed cash buffer enough?

Possibly, but a fixed fraction can be too large or too small relative to dated payments. The sensitivity tool compares it with a payment-linked reserve. This is an illustrative comparison, not a proven optimal reserve policy.

15. Why twelve models if the proposal uses none for forecasts?

The models are research work. Their existence does not create evidence of investable skill. The final strategy should use only components the team can support and explain, not every available implementation. [C01-C03]

16. Has EWMA risk sizing been validated?

No allocation benefit is demonstrated here. EWMA is a transparent risk-monitoring candidate. Finite covariance output and code tests are insufficient to show improved funding or investment outcomes.

17. Why not use M11 for multi-period planning?

The frozen controller fails a basic accounting diagnostic and its legacy evaluation is invalid. We excluded it from the new calculator and retained the defect for a separately amended repair. [C05]

18. Why not use M12 as the integrated solution?

Its legacy runner evaluates part of the market fit in sample and integrates an unsupported controller. No defensible end-to-end benefit is established. Integration alone is not evidence of suitability.

19. What would make you change the strategy?

A verified change in client constraints, security eligibility, expected payment timing or a documented risk breach would trigger review. Numerical triggers cannot be finalized without the case. A short run of favourable returns is not enough.

20. What is the revenue model?

This is an investment competition package, not a venture. No fees, customers, revenue or market size are claimed. The relevant economics are costs, risk, inflation and the client's funding needs.

Research integrity

21. What exactly invalidated the apparent forecasting win?

The synthetic features contained the contemporaneous driver of the label, so the model used information unavailable at prediction time. The resulting score is retained as a leakage demonstration, not forecasting evidence. [C02]

22. Why should we trust anything after that mistake?

Trust should depend on inspectable data paths, regression tests, retained adverse findings and reproducible artifacts. Same-machine replay increases confidence in repeatability, but does not replace an independent audit. [C04, C08]

23. Was the lockbox reopened to get a better result?

No lockbox run was performed in this pass. Frozen artifacts and model files were hash-checked against the starting snapshot. The existing repaired study is explicitly post-lockbox exploratory. [C07]

24. Is the ETF target definitely SPY?

No. The cached asset-0 target is inferred to be AGG from the historical download ordering and signatures. The manifest labels the mapping INFERRED. The filename alone is not proof of column identity.

25. Does no significant result prove no predictability exists?

No. It means the tested study did not establish the claimed advantage under its protocol and sample. Other methods, horizons or assets could differ; they require their own valid evidence. [C03]

26. Why is beating persistence insufficient?

For approximately unpredictable returns, yesterday's return can be a weak forecast. Shrinking toward zero may beat it without adding useful information. The historical mean and zero forecasts are stronger sanity checks in this setting.

27. What was the primary comparator?

The repaired study declares standardized ridge versus the historical mean as its primary contrast. Its broader challenger comparisons include zero and persistence alternatives, with multiplicity handling recorded in the result artifact. [C03]

28. Are folds and seeds independent evidence?

Not automatically. The legacy folds collapsed to the same split and ETF seeds read the same data. The repaired study uses disjoint test windows, but temporal dependence and shared training still limit independence. [C01, C03]

29. Why not report the best model from the old matrix?

Those scores were generated under leakage and degenerate splits, and some models measured different targets. Selecting the smallest pooled score would not establish a valid winner. [C01]

30. How many of the old multiplicity tests support a claim?

The legacy correction found zero of ten comparisons testable after duplicate handling. That is a data-design failure, not ten valid null discoveries or evidence that every model is equivalent.

31. Did the new replay change the findings?

It matched all 68 saved contrasts within the recorded numerical tolerance. It did not create a new independent sample or convert exploratory results into confirmation. [C04]

32. Why are some synthetic results near perfect?

The legacy-feature leakage demonstration is designed to exhibit the invalid information path. It must remain visibly separated from causal forecasts. Its high score is evidence of the defect. [C02]

33. Does the synthetic world demonstrate economic realism?

No. The repaired SIGNAL control has an i.i.d. latent, so lagged data are not expected to recover that contemporaneous signal. It is useful for catching spurious predictability, not validating market realism.

34. What would support a confirmatory claim?

A new fixed protocol with causal features, correct target identity, disjoint evaluation, strong comparators, costs where relevant, multiplicity control and genuinely untouched data. The spent lockbox cannot be repurposed.

35. Can another team reproduce the study from git alone?

Not the exact ETF numbers: the historical cache is ignored by git and has a recorded hash. Downloading again may change the data. Cache availability and redistribution terms remain unresolved limitations.

Financial stress and accounting

36. Where did the return assumptions come from?

They are explicit modelling choices in the JSON config, not estimates fitted to market data. Each scenario is labelled assumption-only. They should be challenged rather than quoted as expected returns. [C06]

37. Why include an extreme joint crash?

It tests what simultaneous losses do to funding and wealth. Its chosen size is not a calibrated historical quantile, probability or prediction. We retain it alongside less adverse paths.

38. Why compare early and late losses?

Payment timing can make the order of returns matter. The two paths use the same shock and other return assumptions to expose that sensitivity. They do not establish a universal ranking of portfolio rules.

39. Does the reserve know future inflation?

It uses the fixed planning inflation assumption and the currently known payment amount. It does not see future realized returns or the realized future inflation path. High-inflation scenarios therefore test a planning mismatch.

40. Are obligations deducted more than once?

Not in the new calculator: each current obligation enters the due amount once, with prior unpaid arrears carried forward. Tests check wealth conservation and the single deduction. The frozen M11 defect is documented separately. [C05, C08]

41. How are trading costs charged?

Costs apply to absolute risky-asset purchases and sales, including proportional liquidation when cash cannot cover a payment. Rebalancing solves for post-cost wealth, preventing an implicit cash injection.

42. Are the costs actual WInS costs?

No. The five cost levels are sensitivity assumptions inherited from the research grid. Actual simulator rules and real-world execution costs must be verified separately.

43. Is the calculator self-financing?

Yes within its stated mechanics: it starts with normalized wealth, earns specified returns, pays costs and obligations, and never adds borrowing or contributions. Unit tests verify the conservation identity under zero returns. [C08]

44. What happens when assets are exhausted?

Wealth cannot become negative; unpaid obligations remain as arrears. The output retains the first shortfall and ending unpaid amount. Clipping portfolio value to zero does not erase the failure.

45. Why report real terminal wealth?

Nominal remaining capital can conceal lost purchasing power. The calculator deflates by the scenario's assumed inflation. That metric is conditional on the assumption, not a measured client outcome.

46. Does drawdown count planned withdrawals as losses?

The reported net investment drawdown excludes external payments through a unit-value calculation. Terminal wealth includes their impact. Both are shown because they answer different questions.

47. Is the reserve always the best policy?

No superiority claim is made. It changes market exposure, liquidity and cost. A fully funded path can still lose purchasing power, and no reserve rescues every infeasible spending plan.

48. Why not report confidence intervals?

The new stress paths are deterministic, deliberately chosen assumptions without estimated probabilities. Confidence intervals would imply a sampling model we have not justified. The forecasting study has separate statistical tests.

49. Are 75 combinations 75 experiments on independent data?

No. They are five assumed paths crossed with three policies and five cost settings. Correlated sensitivity cells do not provide independent replication. [C06]

50. What important costs or features are omitted?

Taxes, asset-specific expenses, cash yield, settlement delay, liquidity impact, external contributions and security eligibility are not modelled. The simplified calculator cannot substitute for a client-specific portfolio simulation.

Competition and execution

51. Is there any real WInS performance in the package?

No verified WInS record is present. Hypothetical calculations and research backtests must never be described as executed competition performance.

52. Where are the original trade notes?

They must come from the actual WInS records, with the original text and execution evidence. No notes are backfilled or fabricated. The recovered instructions confirm three executed notes and a reflection of at most 100 words each. [S06]

53. Have you verified the exact final-report rubric?

Public criteria and recovered IPS/Trading Notes requirements are verified. The final-report instructions and numerical judging weights remain missing. [S01-S06]

54. Is the pitch five minutes because the rules require it?

No. Five minutes is an internal rehearsal plan. Official semifinal or final timing is unverified and must be taken from the team instructions.

55. Why is the package not submission-ready?

Complete client facts, final-report instructions, eligibility, executed trade evidence and student review are missing. Explicit submission authority is also absent. Completed local tools cannot satisfy those external requirements.

56. Does portfolio return determine the winner?

The public competition description emphasizes strategy quality, client fit, research and communication. We cannot infer a numerical judging formula from that description. [S01]

57. Was AI used?

Yes. Codex assisted with code review, calculation tools, source verification and these drafts. Students must verify and attribute retained material and cannot present it as their own independent work. [S02]

58. What did the students personally decide or learn?

No student decision or experience has been documented in this workspace. The final account must come from the students' actual process. This draft does not invent first-person teamwork stories.

59. What is the immediate next action?

Obtain the current registered-team Apply materials and extract source-backed facts. That enables a real client-specific policy and validation against the actual deliverable instructions.

60. What is the strongest honest claim today?

The repository has a reproducible exploratory baseline result, preserved negative findings, a tested assumption-only cash-flow calculator and a structured preparation package. It has not established an investment edge, client suitability, independent scientific validation or competition submission readiness.

Evidence appendix and source register

AI-assisted internal reference pack; reviewed 2026-09-27. File references are repository-relative. Historical evidence is preserved; new calculator evidence is a separate generation.

Public official sources

ID	Source	Verified scope
S01	Wharton competition	2026-27 dates and deliverable names; strategy, client alignment, research and communication guide evaluation
S02	Rules and roles	Team and advisor conditions; no client contact; student ownership and AI attribution; detailed instructions are team-gated
S03	Official FAQ	Competition runs September 28 through December 4, 2026; current materials are distributed through Apply
S04	Official Apply portal	Location of private team materials, not a source for client facts available in this workspace

Recovered PDFs now verify IPS and Trading Notes formatting and contents; see S05-S06 below. Numerical rubric weights, the complete client case, security eligibility and minimum overall trading activity remain unverified. Older-season pages and the prior client are not substitutes. Public notes are retrieval summaries, not archived copies of the private rulebook.

Recovered local sources

S05: data/source_documents/2026_WGY_Investment Policy-FINAL.pdf, pages 1-3: objectives, strategy permanence, 50/500 word limits, title plus two strategy pages, Times New Roman 12-point double spacing, one-inch margins, no formal citations or graphics, at most 5 MB.

S06: data/source_documents/2026_WGY_Trading Notes Analysis-FINAL.pdf, pages 1-2: Laura, operating/facility context, three exact notes from executed trades and at most 100 words per reflection, submitted via Apply form. Both sources have hashes in protocol/RECOVERED_DOCUMENTS.json; they were not reauthenticated with Apply.

Claim-to-artifact map

Claim ID	Permitted statement	Evidence	Limit
C01	Legacy matrix executed 1181 manifest cells	protocol/EXPERIMENT_MANIFEST.csv	Leakage-contaminated; folds/seeds do not establish independent replication
C02	Synthetic lockbox forecasting claim withdrawn	RESEARCH_AUDIT.md; protocol/DECISIONS.md D-050	Do not reuse it as an alpha claim
C03	Repaired tested baselines show no statistically supported mean-benchmark edge	runs/repaired/REPAIRED_RESULTS.json	Exploratory; one-step asset-0 evaluation, not all models or assets
C04	Same-machine replay matched 68 saved contrasts	audit/astra_2026-09-27/VERIFICATION.json	Repeatability, not independent validation
C05	Frozen M11 fails a simple cash-path diagnostic	same verification receipt; D-059	Not a repaired controller or validated allocation
C06	75 policy/path/cost combinations calculated	runs/competition_stress/RESULTS.json and SUMMARY.csv	Deterministic assumptions; not 75 independent observations
C07	All historical protected files remained byte-identical at verification	audit/astra_2026-09-27/START_STATE.json and VERIFICATION.json	Source snapshot check only
C08	Local tests pass	audit/astra_2026-09-27/TESTS.txt	Engineering check, not proof of forecasting or funding performance
C09	Working package contains 60 judge questions	JUDGE_QA.md	Preparation content, not official judge questions

Every calculator number is generated from the config and source hashes recorded inside RESULTS.json. Monthly ledger: wealth_start, obligation, due_with_arrears, paid, arrears, cash, wealth_end, fees, turnover, net_return and unit_value. Units are fractions of initial wealth. Monthly net drawdown excludes withdrawals; terminal wealth includes them. An overdue payment is not added twice to total contractual obligations.

Assumption register

Input	Value / location	Classification
Initial wealth	1.0	Normalization, not dollars or actual client wealth
Horizon	120 months	Modelling choice, not client horizon
Base monthly payment	0.002 of initial wealth	Modelling choice; inflation-linked within each assumed path
Steady annual sleeve returns	4% / 2%	Assumptions; not estimated expected returns
Joint one-month shock	-40% / -15%	Stress assumption, applied early or late
Reserve horizon	12 months, capped at remaining horizon	Policy choice, not optimized
Planning inflation	2.5%	Fixed assumption available to the policy
Realized inflation	2.5% or 8%	Scenario assumptions; not inflation forecasts
Trading cost	0/5/10/25/50 bps per risky dollar traded	Sensitivity grid; no claim about WInS fees
Cash return, contributions, tax	Zero / none	Simplifications
Scenario probabilities	None assigned	No estimated failure probability or confidence interval

Evidence still required

The full official client case, final-report rubric/instructions, registration and eligibility, numerical client facts, current allowed-security list, verified executions and original trading notes are missing. IPS and Trading Notes instructions are recovered. No customer interviews, commercial revenue or partnerships are claimed; these are not applicable to the investment competition brief. The cached ETF's column mapping remains inferred, and redistribution rights are unresolved. A genuinely confirmatory forecasting result would need a new protocol and untouched data. No paper-ready or investment-ready status is asserted.

AI assistance record

On 2026-09-27, Codex assisted with public-rule verification, code inspection, calculator development, local testing, and preparation of these working documents. Students must verify sources, make their own investment decisions and describe only their real work and experience. Material retained from this package must be acknowledged under the official AI policy. No student review or advisor approval has been recorded.